

Sarthak Consul

☎ (+91) 9987104320 • ✉ sarthakconsul@iitb.ac.in • 🌐 sconsul.github.io

Education

Indian Institute of Technology Bombay, Mumbai, India
Bachelor of Technology (with Honours) in Electrical Engineering

July 2016 - Present

- **Major GPA:** 9.62/10
- **Minor** in Computer Science and Engineering

Research Internships & Projects

Segmentation of Lacunar Objects from Ultra High Resolution μ CT Bone Scans

ETH Zürich

Guide: *Prof. Dr. Ralph Müller, D-HEST, Institute for Biomechanics*

Summer 2019

- Worked on extracting lacunar structures from trabecular bone samples of rare disease patients
- Modified Fully Convolutional Networks like WNet for segmentation in 3D images in an unsupervised setting
- Worked on a 2 stage method that extracts regions using adaptive thresholding followed by a classifier to fine-tune results by removing noise structures, achieving an **F1 Score** of **98.4%**

Semantic Segmentation of Medical Images

IIT Bombay

Guide: *Prof. Amit Sethi, EE, Medical Deep learning and AI Lab*

Summer 2018

- Tested ability of Fully Connected Convolutional Networks in segmentation tasks in medical images.
- Implemented Global Convolution on a fully-connected ResNet backbone using Pytorch for the **segmentation of lungs** from chest x-rays and **nuclei segmentation** from h/he stained histopathological sections of organs
- Part of a five member team testing various architectures and conventional image processing techniques as post-processing to improve the state of the art results of **nuclei segmentation** to a mean F1 score of **85%**
- Reviewed research papers on **ReNet** and **ReSeg**, **MobileNet**, and **Siamese Neural Networks**
- **Mentored** two freshmen in the team on the basics of machine learning and deep learning
- Work was used to host the MoNuSeg challenge at MICCAI 2018.
- Annotated and pre-processed data from around 800 patients affected by oral cancer

Scholastic Achievements

- Awarded **Advanced Performer** grade (AP) in **4 courses**: *Biology* (15 out of 445), *Introduction to Electrical Systems* (2 out of 149), *Network Theory* (10 out of 143), and *Electronic Devices Lab* (3 out of 139)
- Granted a **Change of Branch** based on excellent academic standing in freshmen year [2016]
- Secured All India Rank 840 (out of around 200,000 candidates) in JEE Advanced [2016]
- Recipient of KVPY Fellowship (**AIR 16**) by Dept. of Science and Technology, Govt. of India [2014]
- Placed in **National Top 1%** in National Standard Examination in Junior Sciences [2012]
- Ranked **4th** in State Science Talent Search Examination (SSTSE), conducted by Govt. of Rajasthan [2014]
- Recipient of **NTSE** Scholarship by National Council of Research and Training [2012]

Miscellaneous Projects

Digitally Programmable Analog Computer

EE344: Electronics Design Lab

Prof. *Mukul Chandorkar*, EE, IIT Bombay

Spring 2019

Designed and developed a standalone digitally programmable analog computer, capable of solving non-linear dynamical systems upto the fifth order, demonstrating hardware-in-loop capabilities. The system is capable of working in real-time, fully programmable and features on-board power management [Technical Report]

Human Detection from Post-disaster LWIR Imagery

EE338: Digital Signal Processing

Prof. *Vikram M. Gadre*, EE, IIT Bombay

Spring 2019

- Designed and implemented a system to detect humans from low-wavelength infrared (LWIR) images for the purposes of streamlining rescue operations
- Used a two-stage approach of extracting regions using **MSER**, followed by using **Integral Channel Features** such as local sums, **Histogram of Gradients** along with the **DCT** as features to classify using a **SVM**.
- Presented to students and teachers of various tier 2 and 3 colleges across the country as part of TEQIP-KIT-2019 workshop under the initiative of the Ministry of Human Resource Development (MHRD), Govt. of India

Audio Source Separation using a Low Latency Neural Network

CS419: Machine Learning

Prof. *Sunita Sarawagi*, CSE, IIT Bombay

Fall 2018

- Developed a novel low-latency neural network to separate the different musical instruments from music
- Used depthwise separable convolutions, inspired by the **GCN** architecture to reduce parameters.

Tree based Stereo Matching for Postcapture Refocus

Prof. Suyash Awate & Prof. Ajit Rajwade, CSE, IIT Bombay

CS663: Digital Image Processing

Fall 2018

- Implemented a Hybrid tree based stereo matching algorithm which calculates the depth of images using Minimum spanning trees.
- Incorporated CLMF (Cross Local Multipoint filter) to use as a edge preserving smoothener for refining the depth map used the depth estimates to interactively refocus and generate depth blurring.

Multicycled and Pipelined Implementation of IITB-RISC ISA

Prof. Virendra Singh, EE, IIT Bombay

EE309: Microprocessors

Fall 2018

- Designed and implemented a 6-stage pipelined multicycle RISC processor in VHDL, consisting of arithmetic, logical and branching instructions, and tested on Altera DE0-Nano FPGA board.
- Implemented Branch Predictors, Priority Encoders and Hazard Detection Units to reduce latencies and significantly reduce the branch hazards.

Analog Sound Locator

Prof. Siddharth Tallur, EE, IIT Bombay

EE230: Analog Systems Lab

Spring 2018

- Designed and implemented an analog circuit to locate distant sound sources. The setup mimicked bi-aural mammalian ears, using the phase difference of the received sound signal to locate the source.
- Amplitude invariance** was achieved using JFET delay circuits with dynamic phase locking
- Circuit was compact and had a fast convergence (spatial angle was obtained **within a millisecond**)
- One of the **winning projects** in the final evaluation (*amongst 68 teams*)

Reaction Game

Prof. Madhav P. Desai, EE, IIT Bombay

EE224: Digital Systems Lab

Spring 2018

- Built a game to measure reaction times to visual stimulus using a Krypton (Altera MAX V) CPLD board
- Modelled the game as a **finite-state machine** and realised it with **register-transfer level** abstraction in VHDL
- Incorporated false-react detection, timeouts and debouncing features. Used the **Double Dabble Algorithm** to convert the binary reaction times to decimal. Integrated a LCD controller to give real-time scores on a LCD

Pulse Rate Sensor Design

Prof. Siddharth Tallur, EE, IIT Bombay

EE236: Electrical Devices Lab

Autumn 2017

- Created a pulse-rate monitor circuit - variation of blood flow is observed by change in reflectivity of a finger
- Detection was done with the help of a PT333 phototransistor (paired with an IR LED). The input signal is sent through a Op-Amp based filtering scheme (band pass filter with a comparator) to minimize noise.

Building a Cryptosuite

Prof. Bernard L. Menezes, CSE, IIT Bombay

CS 101: Computer Programming

Autumn 2016

- Designed a program that encrypted and decrypted alphanumeric messages using **RSA Algorithm**
- Computed Discrete logarithms using **Baby Step Giant Step Algorithm**, binary and prime field operations, and modulo arithmetic (*addition, multiplication, exponentiation by binary exponentiation, and inverse*)

Mentorship Experience

Teaching Assistant

Prof. Swapneel Mahajan, MA, IIT Bombay

MA207: Partial Differential Equations

Fall 2018

- Taught and held doubt-clearing sessions for a class of approximately 60 students. Involved in the formulation of examinations, grading of answer scripts and other logistics.

Instructor of Machine Learning Bootcamp

Summer 2018

- Taught the fundamentals of Machine Learning to a class of approximately **130** students
- Covered the basics of ML before delving into the theory and implementation of popular algorithms of supervised and unsupervised learning such as **Random Forest Algorithm**, **CNNs**, and **k-means Clustering**.
- Prepared Jupyter notebooks (in Python), meant to serve as tutorials, for the various algorithms discussed

Department Academic Mentor

2019-20

Mentoring a second year student with a view to help him cope academically

Technical Skills

Programming Languages - C, C++, MATLAB, Python, Bash, SQL, VHDL, Assembly and Embedded C

Software libraries - PyTorch, TensorFlow, Keras

Web Development - HTML5, CSS3, PHP

Simulation Tools - NgSpice, Simulink, LabVIEW, Wireshark, ns-3

Design Software - EAGLE, AutoCAD, SolidWorks, Photoshop, L^AT_EX

Hardware - Arduino, Intel 8081, Intel 8085, Altera DE0-Nano

Relevant Coursework

Electrical Engineering

Microprocessors, EM Waves, Probability and Random Processes, Network Theory, Microelectronics, Analog and Digital Systems, Digital Signal Processing, Digital and Analog Communication, Control Systems, Wavelets**

Mathematics Courses

Matrix Computations, Linear Algebra, Data Analysis, Calculus, Differential Equations, Complex Analysis

Computer Science and Engineering

Advanced Machine Learning*, Reinforcement Learning*, Automatic Speech Recognition*, Computer Vision, Machine Learning, Digital Image Processing, Medical Image Processing**, Advanced Image Processing**, Computer Networks, Data Structures and Algorithms, Operating Systems, Logic for CS*

Online Courses

- Reinforcement Learning (DeepMind)
- Convolutional Neural Network for Computer Vision (Stanford University - CS231n)
- Machine Learning (Stanford University - CS229 and Coursera)
- Advanced MATLAB for Scientific Computing (Stanford University - Lagunita) [with Distinction]
- Neural Networks for Machine Learning (University of Toronto - Coursera)

* Indicates courses to be completed by Dec 2019

** Indicates courses to be completed by May 2020

References

Prof. Amit Sethi

Dept. of Electrical Engineering
Indian Institute of Technology, Bombay

Prof. Suyash Awate

Dept. of Computer Science & Engineering
Indian Institute of Technology, Bombay

Prof. Vikram M. Gadre

Dept. of Electrical Engineering
Indian Institute of Technology, Bombay

Prof. Dr. Ralph Müller

Dept. of Health Sciences and Technology
ETH Zürich

Prof. Swapneel Mahajan

Dept. of Mathematics
Indian Institute of Technology, Bombay